Predicting Customer Holiday Booking

Machine Learning Model Random Forest Classifier

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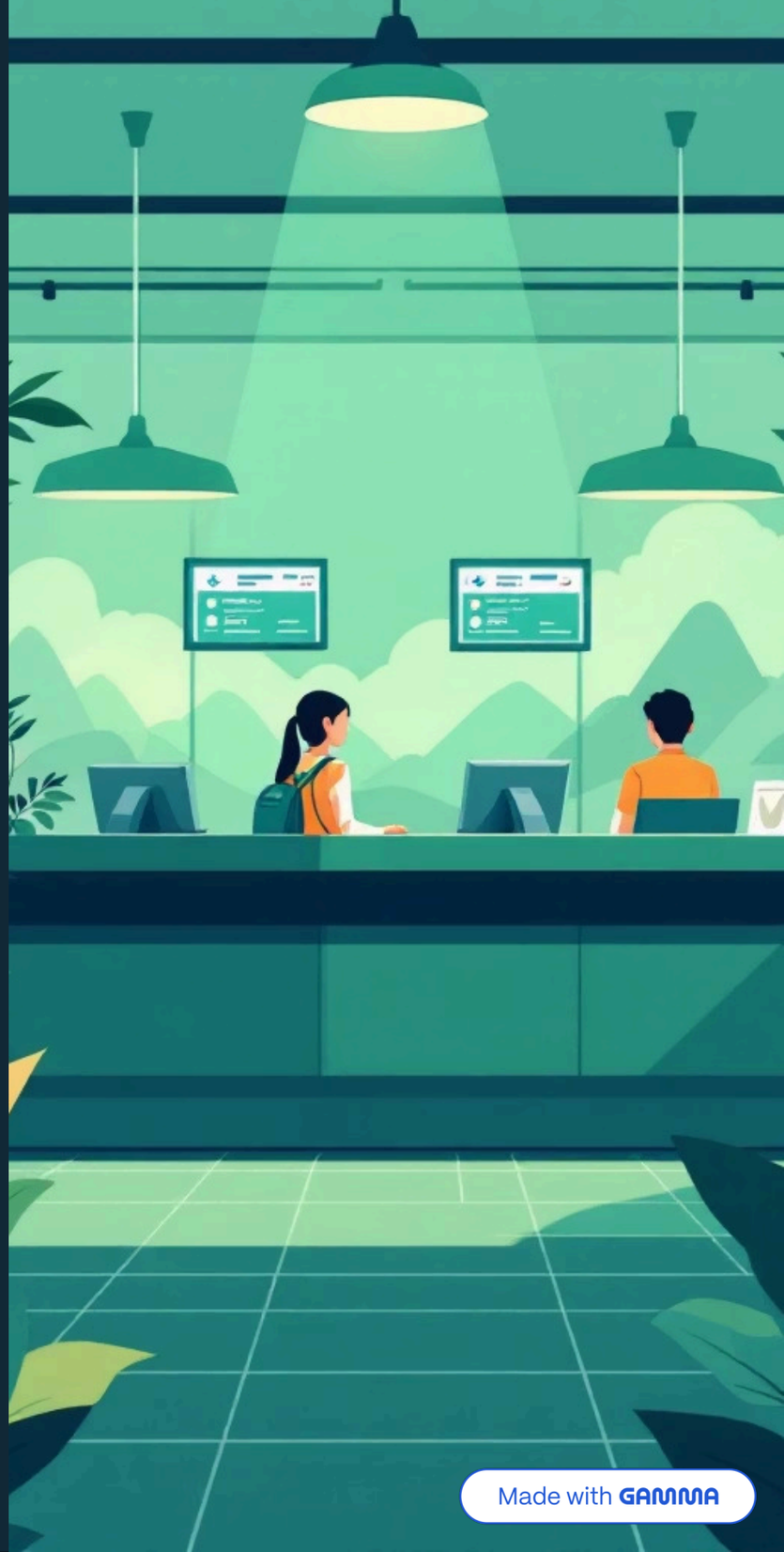
1-March-2026



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Business Objective

Strategic Goal

Customers rarely purchase flights at the airport. Our aim is to proactively identify which customers are likely to complete a booking before they reach that point.

Model Purpose

Build a predictive model that identifies booking behaviour early, improves targeted marketing campaigns, and increases overall conversion rates across the customer journey.

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Dataset Overview

10K
Customer Records
Complete dataset size

14+
Features
Booking-related variables

8,520
No Booking
Class 0 instances

1,480
Completed Booking
Class 1 instances

Target Variable

Binary indicator: **booking_complete** (0/1)

Key Variables

- Purchase lead time
- Flight duration
- Booking origin
- Add-ons (baggage, meals, seat)
- Trip type
- Flight day

Note: Dataset is imbalanced with more non-bookings than completed bookings

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Machine Learning Model

Why Random Forest?

- Handles both categorical and numerical data effectively
- Performs well with imbalanced datasets
- Provides interpretable feature importance
- Reduces overfitting through ensemble approach

Model Configuration

Train/Test Split: 80% training, 20% testing

Target: Binary classification (booking completion)

Random Forest Classifier

An ensemble learning method that builds multiple decision trees and combines their predictions for improved accuracy and robustness.

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Model Performance Metrics

Overall accuracy of 85% is misleading due to class imbalance. Class-wise breakdown reveals performance disparities.

```
[[8336  184]
 [1286  194]]
              precision    recall  f1-score   support

           0       0.87      0.98      0.92      8520
           1       0.51      0.13      0.21      1480

    accuracy                           0.85     10000
   macro avg       0.69      0.55      0.56     10000
weighted avg       0.81      0.85      0.81     10000
```

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Confusion Matrix Analysis

True Negatives: 8,336

Correctly predicted non-bookings

False Positives: 184

Incorrectly predicted bookings

False Negatives: 1,286

Missed actual bookings

True Positives: 194

Correctly predicted bookings

Key Insight

Model predicts non-bookings very well (98% recall) but struggles to detect actual bookings. Only 13% recall for booking class means the model misses many real customers who actually book.

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Feature Importance

Most Influential Variables

01

Purchase Lead Time

Days between booking and flight

02

Flight Duration

Length of the journey

03

Length of Stay

Duration at destination

04

Extra Baggage

Additional luggage selected

05

Booking Origin

Source of booking request

06

Add-on Services

Meals, seats, insurance options

Interpretation

Customers who book earlier, choose add-ons, and have longer stays are more likely to complete their booking. These features drive predictive power.

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Key Business Insights

Purchase Behaviour Matters

Early planners show significantly higher booking probability. Customers researching flights well in advance demonstrate stronger purchase intent.

Add-ons Indicate Intent

Customers selecting extra baggage, in-flight meals, or preferred seats are more likely to convert to completed bookings.

Model Bias Warning

Model favours predicting "No Booking" due to data imbalance. This means many potential bookings may be overlooked in current predictions.

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Model Limitations

Dataset Imbalance

Significant class imbalance (85% non-bookings) skews model predictions towards majority class, reducing sensitivity to booking behaviour.

Low Recall for Bookings

Only 13% recall for booking class means model misses 87% of actual bookings. This limits practical marketing value.

Feature Engineering Challenges

Feature importance plot is crowded due to one-hot encoding of categorical variables, making interpretation difficult.

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Recommendations

Short-Term Actions

- Use model to filter high-probability leads for targeted campaigns
- Prioritise customers selecting add-ons in marketing outreach
- Offer personalised promotions to early planners

Model Improvements

- Apply SMOTE to balance dataset
- Tune hyperparameters for better recall
- Try Gradient Boosting or XGBoost
- Optimise for Recall instead of Accuracy

Next Steps: Collaborate with marketing team to design A/B tests for personalised campaigns using high-probability segments identified by the model.

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